



Indywidualny Plan Badawczy (IPB) Individual Research Plan (IRP)

1. Podstawowe dane / Basic data

Podstawowe informacje dotyczące doktoranta i realizowanej rozprawy doktorskiej. W przypadku promotora spoza Politechniki Wrocławskiej należy podać afiliację. / *Basic information on the doctoral student and the doctoral dissertation underway. In the case of a supervisor from outside Wrocław University of Science and Technology, affiliation should be provided.*

Imię i nazwisko / Name, surname	Grzegorz Piotrowski
Dyscyplina kształcenia / Discipline of education	informatyka techniczna i telekomunikacja / information and communication technology
Wydział / Faculty	W4 Wydział Informatyki i Telekomunikacji / Faculty of Information and Communication Technology
Katedra / Department	K46 Katedra Sztucznej Inteligencji / Department of Artificial Intelligence
Data rozpoczęcia kształcenia / Date of commencement of the doctoral education	01.10.2025
Nr ORCID	0009-0004-6013-0461
Promotor/ promotorzy / Supervisor(s)	dr hab. inż. Tomasz Jan Kajdanowicz, prof. uczelni, Politechnika Wrocławska (K46, W4), ORCID 0000-0002-8417-1012
Promotor pomocniczy / Assistant supervisor	—

2. Temat rozprawy doktorskiej / Topic of doctoral dissertation

Propozycja tematu rozprawy doktorskiej. Przedstawiony temat nie musi stanowić ostatecznego tytułu, jaki zostanie nadany rozprawie doktorskiej złożonej na zakończenie kształcenia w Szkole Doktorskiej. / *Proposed topic for a doctoral dissertation. The presented topic does not have to constitute the final title of the doctoral dissertation submitted at the end of education at the Doctoral School.*

PL: Metoda dostrajania modeli wizja–język–działanie w pętli rzeczywistość–symulacja–rzeczywistość wymagająca mniej danych rzeczywistych: ukierunkowane na zadanie zbieranie danych do budowy cyfrowych bliźniaków, uczenie w bliźniakach i modelach świata oraz aktywny wybór nielicznych danych rzeczywistych

EN: A method for fine-tuning vision-language-action models in the real-to-sim-to-real loop that needs less real data: task-aware capture for building digital twins, learning in twins and world models, and active selection of few real-world data

3. Harmonogram przygotowania rozprawy doktorskiej / Schedule for the preparation of the doctoral dissertation

Harmonogram przygotowania rozprawy doktorskiej z wyszczególnieniem etapów i miejsc prowadzonych badań, w tym terminów zakończenia badań częściowych (semestr kształcenia) i



opracowania ich wyników oraz opcjonalnie w zależności od rodzaju prowadzonych badań, wybranych między innymi spośród następujących:

- realizacja badań (w tym przygotowanie stanowiska badawczego, przygotowanie materiałów do badań, wykonanie pomiarów/eksperymentów, opracowanie otrzymanych wyników),
- przygotowanie i wysłanie artykułu naukowego do redakcji czasopisma naukowego,
- wysłanie zgłoszenia patentowego do właściwego urzędu,
- opracowanie propozycji wdrożenia dla przemysłu,
- udział w konferencjach, warsztatach lub szkołach letnich/zimowych i prezentacja wyników badań,
- przygotowanie wniosku i wystąpienie o grant,
- mobilność krajowa lub zagraniczna doktoranta (w tym konsultacje naukowe i wizyty studyjne w ośrodkach krajowych lub zagranicznych).

A schedule for the preparation of the doctoral dissertation detailing the stages and locations of the research to be conducted, including the dates for completing the sub-studies and developing their results, and optionally, depending on the type of research to be conducted, selected among the following:

- *execution of the research (including preparation of the research site, preparation of materials for the research, execution of measurements/experiments, elaboration of the obtained results),*
- *preparation and submission of a scientific article to the editor of a scientific journal,*
- *submission of a patent application to the relevant office,*
- *development of an implementation proposal for the industry,*
- *participation in conferences, workshops or summer/winter schools and presentation of research results,*
- *preparation of an application and application for a scholarship,*
- *domestic or foreign mobility of the doctoral student (including scientific consultations and study visits to domestic or foreign centres).*

Okres realizacji zadania (semestr kształcenia) / Task implementation period (semester)	Zwięzły opis zadania / Brief description of the task
Semestr 1	<i>(Oct 2025 – Feb 2026)</i> Research target: define the problem and map the state of the art. T1.1 Systematic literature review in four areas: (a) sim-to-real transfer and domain randomization, (b) learning under distribution shift and domain adaptation, (c) neural scene reconstruction (NeRF, 3D Gaussian Splatting), (d) learning-based visual navigation. <i>D:</i> annotated bibliography and a written research-gap statement (basis of §6). T1.2 Identification of the open problem: how much real data sim-to-real transfer needs, and whether reconstructed scenes can replace real data. <i>D:</i> problem statement and preliminary research questions, discussed with the supervisor. T1.3 Doctoral School courses of semester 1. <i>D:</i> completed courses. Place: K46.



Semestr 2	(Mar – Sep 2026) Research target: formulate the research plan and select the experimental basis. T2.1 Formulation of the data-efficient real-to-sim-to-real thesis, research questions RQ1–RQ4 (capture less, learn robustly in an imperfect twin, few real data, budget of the whole loop) and hypotheses H1–H4 with measurable decision rules and evaluation tiers A–C. <i>D</i>: §7 and §9 of this IPB. T2.2 Comparative survey of public benchmarks and datasets (e.g. ScanNet++, HM3D), simulators (e.g. Habitat-Sim) and 3D Gaussian Splatting implementations against the needs of RQ1–RQ4 (real video + reference scan, licence, compute cost). <i>D</i>: tool-selection note with a shortlist for Stage I. T2.3 Individual Research Plan agreed with the supervisor. <i>D</i>: this IPB, signed and submitted by 30.09.2026. Place: K46.
Semestr 3	(Oct 2026 – Feb 2027) Stage I – method component C1: capture less (RQ1, H1). T3.1 Tier A set-up in both testbeds (§9): navigation, ≥ 20 public scenes (10 held out) with laser-scan references and separate phone captures; manipulation, ≥ 20 simulator task–object configurations (10 held out) with held-out physical parameters. Existing 3DGS pipeline and system identification from the T2.2 shortlist; twins at ≥ 4 capture budgets; open VLA (LoRA fine-tuning), VLM and world-model checkpoints running on the twins (§9); pilot fixing task difficulty (below the P ceiling), the operator-time cost model and the measured GPU cost; WCSS/PLGrid computing grant; pre-registration of the protocol (§7, §9). <i>D</i>: working loop and protocol in the repository for both testbeds, by Jan 2027. Place: K46, WCSS. T3.2 Validation stand: written agreement on robot access (planned cooperation with the K29 Denali laboratory for the mobile robot; a PWr manipulator if available) and sensor decision (SzD Minigrant application if a sensor must be bought). <i>D</i>: agreement, by Nov 2026. T3.3 C1: VLM-guided, task-aware, uncertainty-guided capture of views and interactions vs. uniform, task-blind, risk-weighted and VLM-guided reconstruction-quality selection, judged by the success of the VLA fine-tuned in each twin. <i>D</i>: first budget–success curves (first H1 result), by Feb 2027.



Semestr 4	(Mar – Sep 2027) Stage I completed; Stage II – method component C2: fine-tune in an imperfect twin and a twin-grounded world model (RQ1–RQ2, H1–H2). T4.1 C2: uncertainty-aware LoRA fine-tuning of the VLA (SFT, then RL) in the twin, with uncertainty-driven randomization and sample weighting, vs. uniform domain randomization (H2a); the world model post-trained on twin rollouts and covering the twin's uncertain regions vs. the twin only (H2b); equal capture budgets, both testbeds. <i>D:</i> H1 completed and first H2 result on tier A, by Apr 2027; H2 completed, by Jul 2027. Place: K46, WCSS. T4.2 Prepare and submit article P1 to NeurIPS 2027 (method component C1, H1, and first results of C2, H2, for a VLA fine-tuned in twins, in manipulation and navigation). <i>D:</i> submission, May 2027 (§11). T4.3 Stage III set-up: C3 trial selection (gap predicted from twin, world-model and VLA uncertainty) and correction code on tier A (Jul–Sep 2027). <i>D:</i> first H3 result, by Sep 2027. T4.4 Grants and international activity: NCN PRELUDIUM application; poster at an international summer school; foreign co-author sought for P2; NAWA Bekker application for the sem. 6 visit. <i>D:</i> applications, poster.
Semestr 5	(Oct 2027 – Feb 2028) Mid-term evaluation (Nov 2027): autoreferat (≤ 5 pages, ~mid-Oct 2027) and 15-min presentation. Stage III – method component C3: collect few real data (RQ3, H3). T5.1 C3: active selection of real trials vs. random and twin-only failure-driven selection; correction of the twin (appearance and physical parameters) and the world model, and fine-tuning of the VLA with the selected trials, in both testbeds. <i>D:</i> H3 completed on tier A, by Jan 2028. Place: K46, WCSS. T5.2 First robot validation campaign (tier C, 2 PWR settings, ~8 robot-hours): own captures, twins, the fine-tuned VLA, uniform capture vs. the method. <i>D:</i> validation report. Place: Denali (and the manipulator's lab, if agreed). Without robot access, the campaign moves to sem. 7; H1–H3 are decided on tier A either way. T5.3 Article P2 (method component C3: active selection of few real data and twin correction, RQ3) submitted to ICML 2028 (~late Jan 2028; CVPR 2028, ~Nov 2027, only if H3 is complete by then). If P1 was rejected, its revision goes to CVPR 2028. <i>D:</i> submission(s).



Semestr 6	(Mar – Sep 2028) Stage IV – the whole method in both testbeds (RQ4, H4, the thesis). T6.1 Real-data budget curves of the whole method vs. real-only fine-tuning of the same VLA ("exchange rate") and vs. the strongest existing twin fine-tuning pipeline for VLAs (TwinRL/RialTo-style) on tier A in manipulation and navigation, with C1–C3 unchanged, plus the tier-B check against published paired sim/real evaluations (SIMPLER). <i>D</i>: first-testbed result by Apr 2028; H4 completed, by Sep 2028. Place: K46, WCSS. T6.2 Foreign research visit (3 months, NAWA Bekker; if not funded, a 5–30-day Erasmus+ short-term mobility to a European lab; hosts in §12): joint experiments. <i>D</i>: visit report. T6.3 Article P3 (the whole method against real-only fine-tuning and the existing twin pipeline in both testbeds; working title "budget law") submitted to NeurIPS 2028 (~May 2028); if not ready, to ICLR 2029 (~late Sep 2028). <i>D</i>: submission.
Semestr 7	(Oct 2028 – Feb 2029) Consolidation (RQ1–RQ4). T7.1 Second robot validation campaign (tier C) of the whole method; final analysis of H1–H4 across tiers and testbeds. <i>D</i>: final results. Place: K46, Denali. T7.2 Release of the method's code, experiment scripts and LoRA adapters (where model licences permit). <i>D</i>: public repository. T7.3 Optional article P4 (consolidated study) to ICLR 2029 (~late Sep 2028) or CVPR 2029 (~Nov 2028); start of dissertation writing. <i>D</i>: submission (optional), dissertation outline.
Semestr 8	(Mar – Sep 2029) T8.1 Editing of the doctoral dissertation; final version. <i>D</i>: dissertation submitted by 30.09.2029 (§4). T8.2 Responses to reviews and camera-ready versions of pending articles. <i>D</i>: revised manuscripts.

4. Termin złożenia rozprawy doktorskiej / Date of the doctoral dissertation submission

Planowany termin złożenia rozprawy doktorskiej nie może przekraczać daty zakończenia ósmego semestru kształcenia w Szkole Doktorskiej. / A planned date for submission of the dissertation, which must not exceed the date of completion of the eighth semester of education in the Doctoral School.

30.09.2029

(rekomenrowana data to ostatni dzień z okresu 4 lata kształcenia w Szkole Doktorskiej, np. jeżeli kształcenie rozpoczęło się 1.10.2025, to proszę wpisać 30.09.2029 /

The recommended date, which marks the final day of the four-year doctoral training period. For example, if your training began on October 1st, 2025, please enter September 30th, 2029)

5. Uzasadnienie wyboru tematu rozprawy doktorskiej / Justification for choosing the topic of the doctoral dissertation

Opis zawierający uzasadnienie wyboru tematu rozprawy doktorskiej uwzględniający potencjalne obszary zastosowania planowanych wyników rozprawy doktorskiej (maks. 1 strona, czcionka 11 pt., interlinia 1). / Description containing the justification for the selection of the topic of the doctoral



dissertation, taking into account the potential areas of application of the planned results of the doctoral dissertation (at most 1 page, font 11, line spacing 1).

Modern machine learning is limited less by models than by **data in the target domain**. Robot manipulators, mobile robots and other systems acting in the physical world need much experience, which is slow, expensive and sometimes risky to collect in reality. Pretrained open **vision-language-action (VLA) models**, trained on large robot datasets, have made the starting point much stronger, but they still need data from the target task and place to work there. Training in simulation avoids that cost but opens a **sim-to-real gap**.

Digital twins built from real data narrow this gap. Neural scene reconstruction (3D Gaussian Splatting, Neural Radiance Fields) turns a short real capture into a photorealistic model of the geometry and appearance of a scene, and system identification estimates its physical parameters (masses, friction, actuation) from a few real interactions. Learned **world models** can generate further variations. This gives a **real-to-sim-to-real loop**: collect real data, build a twin, fine-tune in the twin, check and correct with a little more real data, deploy. Building twins, fine-tuning VLAs in simulation and training them in world models are all active research areas. What remains open is the **real data** the whole loop consumes. Each step spends it: the capture that builds the twin, the real data used to make learning robust to what the twin got wrong, and the real-world trials used to check and correct the result. Today these amounts are chosen by habit: data is collected uniformly, learning uses generic randomization, and real trials are picked at random or by hand. How much real data a twin actually saves has, to our knowledge, not been measured systematically across tasks.

Goal of the dissertation. The goal is to develop a **method** for learning in the real-to-sim-to-real loop that reaches a given real-world performance with **significantly less real data** than existing approaches. A pretrained open VLA is fine-tuned sim-first in the twin. The method has three components, one per loop step: (C1) capture guided by a vision-language model (VLM) to the task-relevant parts of the scene where the twin is most uncertain, (C2) parameter-efficient fine-tuning that is robust to the twin's errors, in the twin and in a world model grounded in it that covers what the twin got wrong, and (C3) active selection of the few real-world data that correct the twin, the world model and the VLA. The **uncertainty** of the twin, the world model and the VLA guides all three. The thesis is that the method needs at most a tenth of the real data of fine-tuning the same VLA on real data only, and at least two times less than the strongest existing twin fine-tuning pipeline, in both testbeds.

Why this topic, and why in this discipline. The object is a general, task- and domain-agnostic data-efficient learning methodology (active learning, learning under distribution shift, uncertainty estimation, adaptation of foundation models), not a particular robot, task, simulator or benchmark, which places it in *information and communication technology*. The student builds on existing open models, twin pipelines, simulators and public datasets. The method is tested with the same settings on two testbeds of equal status, **robotic manipulation** and **visual navigation**, where public data make it possible to count real data exactly without an own robot. Trials on a real robot (planned cooperation with a PWr robotics laboratory) validate, but do not decide, the conclusions. The topic fits the Department of Artificial Intelligence (K46), whose research groups include representation learning. Parameter-efficient fine-tuning of open VLAs runs on the GPU infrastructure available to PWr researchers (WCSS, PLGrid).

Potential application areas: faster and cheaper deployment of robot foundation models in new places, such as robot workcells, warehouses, buildings and inspection sites, with less time spent collecting data and supervising real trials; guidance on how much real data to collect for a twin; and, beyond robotics, any model adapted on reconstructed or simulated data and used on real sensor data.

6. Zarys aktualnego stanu badań w tematyce rozprawy doktorskiej / **Outline of the current state of research in the topic of the doctoral dissertation**



Przegląd literaturowy zawierający opis aktualnego stanu badań w tematyce rozprawy doktorskiej uwzględniający odwołania do najważniejszych pozycji literaturowych związanych z tematyką rozprawy doktorskiej (maks. 2 strony, czcionka 11 pt., interlinia 1). / A literature review containing a description of the current state of research in the subject of the doctoral dissertation, including references to the most important literature items related to the subject of the doctoral dissertation (at most 2 pages, font 11, line spacing 1).

Real-to-sim-to-real with digital twins. A real-to-sim-to-real loop builds a simulation of the target scene or object (a *digital twin*) from real data, learns a policy in it and transfers it back. Generic simulators such as Habitat [1] train at scale, but not in the target scene. 3D Gaussian Splatting (3DGS)

[2] turns a short camera capture into a photorealistic twin, and policies trained in such twins transfer in manipulation (RialTo [3], SplatSim [4]) and navigation (EmbodiedSplat [5]); GPU simulators ship twin environments (ManiSkill3 [6]). Physical parameters are inferred from real trajectories as a posterior (BayesSim [7]). Building a twin is no longer the bottleneck; **real data** is: the capture that builds the twin and the real trials that correct it. Current systems fix this cost by hand; RialTo's ablation over 0–15 real demonstrations is the closest published analysis of a real-data budget.

Pretrained vision-language-action (VLA) policies. Open VLA models map camera images and an instruction to actions: OpenVLA [8], $\pi 0$ [9] and, for navigation, NaVILA [10]. They need fine-tuning for a new setup. Low-rank adaptation (LoRA) [11] makes this feasible on a single GPU [8], and the OFT recipe [12] raises success and speed. Fine-tuning still consumes real demonstrations: in navigation, more data from a known location saturates quickly [13], and sim-and-real co-training raises real success by 38% on average [14].

Fine-tuning VLAs in simulation, twins and world models (crowded). Reinforcement learning (RL) fine-tuning of VLAs in simulation improves on supervised fine-tuning, especially under distribution shift [15, 16]. TwinRL [17] fine-tunes a VLA with RL in a twin reconstructed from a smartphone capture and uses the twin to find failure-prone configurations for targeted real rollouts, reporting 20 minutes of on-robot interaction. Learned **world models** serve as the simulator instead: open world foundation models are post-trained per setup (Cosmos [18]; NWM [19] for navigation), VLAs are optimized on-policy inside them without real interaction [20], and VLAW [21] improves a world model with real rollouts and then the VLA with its synthetic data. GigaWorld-0 [22] combines 3DGS reconstruction, system identification and video generation into a data engine for VLAs. Open vision-language models (VLMs) such as Qwen2.5-VL [23] can judge task success [24]. Here they are components and baselines, each reported at a fixed, hand-chosen real-data budget.

Capturing less: active reconstruction and identification. Active view selection chooses the images that improve a reconstruction most: FisherRF [25] maximizes expected information gain on radiance-field parameters, and Bayes' Rays [26] estimates an uncertainty field for a trained radiance field. Risk-aware view acquisition [27] weights FisherRF by safety-critical regions, ASID [28] designs exploration that identifies physical parameters from little real data, and AREA3D [29] adds VLM guidance to active reconstruction. Their objective is reconstruction quality, safe exploration or parameter accuracy, not the success of a policy fine-tuned in the twin; we found at most two task-aware capture papers a year in 2023–2026. ScanNet++ [30] pairs laser scans and DSLR images with a separate phone stream, so budgets can be varied offline.

Learning in an imperfect twin, and collecting few real data. Domain randomization varies appearance

[31] or dynamics [32] uniformly within hand-set ranges; frozen encoders (DINOv2 [33]) make twin and real

samples comparable. A twin's own reconstruction uncertainty is rarely used as a training signal, and we found no world model aimed at the regions where the twin is uncertain. On the real side, prediction-powered inference [34] combines many cheap predictions with a few labels, and SIMPLER [35] shows on paired simulated and real evaluations that simulation can track the real performance of manipulation policies. TwinRL picks failure-prone configurations for the policy, and VLAW corrects



the world model with rollouts that are not selected; neither selects the few real trials by the joint uncertainty of twin, world model and policy under a counted budget.

Research gap. Real-to-sim-to-real fine-tuning of VLAs works, but its real-data cost is set by hand, spent uniformly and measured in one domain per study. Missing is one method that allocates it actively at every step: (i) VLM-guided, task-aware capture of appearance and physical parameters, judged by task success (RQ1); (ii) fine-tuning in the twin and in a world model grounded in the twin, driven by the twin's uncertainty (RQ2); (iii) active selection of the few real data that correct twin, world model and VLA (RQ3); and (iv) evidence that it needs less real data than real-only fine-tuning of the same VLA and the strongest existing twin pipeline, measured the same way in manipulation and navigation (RQ4).

References

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7. Pytania i hipotezy badawcze / Research questions and hypotheses

Opis najważniejszych pytań i hipotez badawczych związanych z realizowaną rozprawą doktorską (maks. 1 strona, czcionka 11 pt., interlinia 1). / Description of the most important research questions and hypotheses related to the doctoral dissertation (at most 1 page, font 11, line spacing 1).



Goal and thesis. The goal is a **method** for learning in the real-to-sim-to-real loop that reaches a given real-world performance with **significantly less real data** than existing approaches: a pretrained open **vision-language-action (VLA) model** fine-tuned sim-first in a digital twin, with one component per loop step: (C1) capture guided by a vision-language model (VLM) to the task-relevant parts of the twin where it is most uncertain; (C2) uncertainty-aware, parameter-efficient fine-tuning in the twin and in a **world model grounded in the twin** that covers the twin's uncertain regions; (C3) active selection of a few real data to correct twin, world model and VLA. Thesis = H4; H1–H3 = component ablations. Testbeds (equal status): robotic manipulation and visual navigation.

Tier A protocol (proxy reality). In each testbed a **reference from a separate, higher-fidelity source plays "reality"**, and the twin (neural reconstruction + system identification) is built from a subset of a separate, cheaper capture, sharing no data. Navigation: ≥ 20 public indoor scenes (10 held out), laser-scan and DSLR reference, twin from the phone capture. Manipulation: a physics simulator with held-out physical parameters and its own rendering (the weaker proxy; tier B, published paired sim/real evaluations of open robot policies, checks it). Real data (capture = views and interaction samples; trials and demonstrations = episodes in the reference) is summed in one operator-time cost. P = task success rate in the reference. Tier A decides H1–H4; tiers B and C (real robot) report agreement. Tests: one-sided, $\alpha = 0.05$, scene/seed bootstrap, Holm correction, pre-registered.

RQ1 (C1: capture less). How little capture does a twin need, and can the task guide it?

H1. C1 reaches the P of uniform capture with $\geq 40\%$ less capture. *Decision:* on fitted capture– P curves, $C_{\text{task}}(\tau_1) \leq 0.6 \cdot C_{\text{uniform}}(\tau_1)$, and the upper 95% bound of $C_{\text{task}} / C_{\text{recon}}$ (task-blind uncertainty selection) is < 1 ; $\tau_1 = P$ of uniform capture at its largest budget.

RQ2 (C2: learn in an imperfect twin). How can fine-tuning be robust to the twin's errors?

H2. At an equal capture budget, (a) C2 (augmentation and sample weights from the twin's uncertainty and the representation distance to a few held-out real samples) beats uniform domain randomization, and (b) twin + world model beats twin only. *Decision:* (a) mean paired P gain ≥ 10 pp on held-out scenes, lower 95% bound > 0 ; (b) lower 95% bound of the paired gain > 0 ; both at each of ≥ 2 capture budgets.

RQ3 (C3: few real data). Which few real data close the gap?

H3. C3 (trials selected by the gap predicted from twin, world model and VLA uncertainty) reaches the target P with $\geq 50\%$ fewer trials than random selection. *Decision:* $N_{\text{active}}(\tau_3) \leq 0.5 \cdot N_{\text{random}}(\tau_3)$ on fitted trial– P curves, upper 95% bound of the ratio < 1 ; $\tau_3 = P$ of random selection at its largest budget.

RQ4 (the whole method). How much real data does the method need versus the alternatives?

H4 (thesis). In each testbed, with C1–C3 and their hyperparameters unchanged, the method reaches the target P with (a) $\leq 10\%$ of the real data of fine-tuning the same VLA on real data only and (b) $\geq 2\times$ less real data than the strongest existing twin fine-tuning pipeline for VLAs (uniform capture, domain randomization, RL fine-tuning in the twin, random or failure-driven real trials). *Decision:* (a) $B_M(\tau_4) \leq 0.1 \cdot B_{\text{real}}(\tau_4)$, upper 95% bound of the ratio < 0.2 ; (b) $B_M(\tau_5) \leq 0.5 \cdot B_{\text{pipe}}(\tau_5)$, upper 95% bound < 1 ; $\tau_4, \tau_5 = P$ of each comparator at its largest budget; H1–H3 effects keep their sign and tier B agrees in direction.

A hypothesis holds when all its parts pass; a failed one is reported as a curve.

8. Wkład spodziewanych wyników dla rozwoju dyscypliny naukowej / Contribution of the expected results to development of scientific discipline

Krótki opis spodziewanego wkładu wyników rozprawy doktorskiej dla rozwoju dyscypliny naukowej, w której realizowana jest rozprawa doktorska (maks. 1 strona, czcionka 11 pt., interlinia 1). / Brief description of the expected contribution of the results of the doctoral dissertation to the development



of the scientific discipline in which the doctoral dissertation is carried out (at most 1 page, font 11, line spacing 1).

The dissertation is in **information and communication technology** (*informatyka techniczna i telekomunikacja*), area machine learning. Its object is the task- and domain-agnostic **real-to-sim-to-real learning loop**: building a digital twin (appearance, geometry and, where relevant, physical parameters) from limited real data, fine-tuning a policy (here a pretrained VLA) in it and transferring the result back to reality; it develops a method that makes this loop work with **less real data**. Building twins, world models, simulators or benchmarks, training foundation models from scratch, control design and mechanical engineering are not the object of the research; the student uses existing tools, open models and public data.

Key original contribution: the method. To our knowledge, the first method for real-to-sim-to-real learning that treats the real data of the whole loop as **one budget allocated actively at every step** when a pretrained open vision-language-action (VLA) model is fine-tuned sim-first in a digital twin and in a world model grounded in it. It targets a given real-world performance with at most a tenth of the real data of real-only fine-tuning of the same VLA and at least two times less than the strongest existing twin fine-tuning pipeline (RQ4, H4), with the same settings in robotic manipulation and visual navigation. Fine-tuning VLAs in simulation and in world models is not new; spending the loop's real data where it matters is. The three components are parts of this one method, each tested as an ablation at its own step:

1. **C1, capture less (RQ1, H1).** A vision-language model picks the task-relevant objects and regions from the instruction and scene, and capture (views for appearance and geometry, interactions for physical parameters) goes where the twin is most uncertain in them; compared with uniform and task-blind selection.
2. **C2, learn robustly in an imperfect twin (RQ2, H2).** Parameter-efficient reinforcement and imitation fine-tuning with augmentation and sample weights from the twin's uncertainty and the representation distance to a few real samples, and a twin-grounded world model covering the twin's uncertain regions; compared with uniform domain randomization and with the twin alone.
3. **C3, transfer with few real data (RQ3, H3).** Active selection of real-world trials by the twin-to-real gap predicted from the uncertainty of twin, world model and VLA, used to correct all three; compared with random selection.

How the method is evaluated and released (supporting the method, not contributions in their own right). Budget–performance curves of the method against real-only fine-tuning and the existing pipeline, which give the "exchange rate" between twin and real data; a proxy-reality protocol on public data that counts real data exactly without a robot (a separate, higher-fidelity reference as "reality"), applied in each domain; and open-source code of the method, built on existing open models, twin pipelines and simulators.

Significance for the discipline. Real target-domain data is the main cost of deploying learned systems; methods that decide where it is worth spending, and a measured estimate of how much of it a twin can replace, are relevant to active learning, learning under distribution shift, uncertainty estimation and adaptation of foundation models in general, beyond robotics.

Dissemination. Results are planned for peer-reviewed conferences worth 200 points and assigned to ITiT on the ministerial list of 5.01.2024: NeurIPS, ICML and ICLR (machine learning) and CVPR (computer vision), with ICCV and ECCV for resubmissions. P1 (C1 and first C2 results, RQ1–RQ2) targets NeurIPS 2027; P2 (C3, RQ3) ICML 2028 or CVPR 2028; P3 (the whole method against real-only fine-tuning and the existing pipeline in both domains, RQ4) NeurIPS 2028, with ICLR 2029 as the fallback.

9. Planowane metody badawcze / Planned research methods



Krótki opis metod badawczych, które są będą użyte w czasie realizacji rozprawy doktorskiej dla (maks. 2 strony, czcionka 11 pt., interlinia 1). / Brief description of the research methods that will be used during the implementation of the doctoral dissertation (at most 2 pages, font 11, line spacing 1).

The dissertation develops **one method** for the loop: real data → twin → sim-first fine-tuning of a pretrained VLA in the twin and in a world model grounded in it → a few real trials → correction of twin, world model and VLA → deployment. Stages I–III develop and ablate its components C1–C3 (RQ1–RQ3), and Stage IV tests the whole method (RQ4, the thesis H4). The twin has two parts: geometry and appearance from existing open-source 3D Gaussian Splatting (3DGS) [2] pipelines, and physical parameters identified from real interactions as a posterior (BayesSim-type [7]). No simulator, benchmark, VLA or world model is trained from scratch.

Models (open weights, adapted). VLA: OpenVLA-OFT [8, 12] for manipulation, with π_0 [9] (openpi) as a second backbone, and NaVILA [10] for navigation, whose mid-level actions ("move forward 75 cm") map onto Habitat actions. All VLA fine-tuning is parameter-efficient: LoRA [11] adapters, supervised fine-tuning (SFT) and then PPO, which generalizes best for VLAs [16], with open RL tooling [15]. VLM: Qwen2.5-VL-7B [23], zero-shot, grounds the instruction to task-relevant objects and regions (C1) and judges success

[24] where the simulator gives no ground truth (checked against it where it does). *World model:* Cosmos-Predict2 [18] (manipulation) and NWM [19] (navigation) checkpoints, post-trained on twin rollouts. *Compute:* published needs are $\geq \sim 27$ GB of GPU memory for OpenVLA LoRA, > 22.5 GB for π_0 LoRA, and one 8-GPU node for RL of a VLA in simulation [15]. We estimate $\sim 23k$ H100-hours for sem. 3–7 (~ 5 H100 on average): WCSS Lem and a PLGrid grant, applied for in T3.1; small VLAs are used for development.

Tier A protocol: proxy reality (decides the hypotheses). In each testbed "reality" is a reference from a separate, higher-fidelity source, and the twin is built from a subset of a separate, cheaper capture. *Navigation:* on ≥ 20 ScanNet++ [30] scenes (10 held out from all tuning), the reference is the laser-scan mesh textured from the DSLR images and rendered in Habitat [1] with a robot-camera model; the twin is 3DGS from a subset of the phone stream, run as a mesh (as in EmbodiedSplat [5]) or with a 3DGS renderer; its physical part is actuation noise. Tasks: instruction and image-goal navigation with a minimum geodesic distance set in a pilot so that P stays below its ceiling. *Manipulation:* ManiSkill3 [6] tasks (≥ 20 task-object configurations, 10 held out); reality is the simulator with held-out physical parameters and its own rendering; the twin is 3DGS from a subset of its views plus parameters identified from a few recorded interactions. Physics is shared, so this is the weaker proxy. *Both:* capture = views and interaction samples; trials and demonstrations = episodes in the reference; all summed in one operator-time cost; the largest budget stays below the full capture. In navigation the proxy's error to held-out real images is reported, and each hypothesis is re-checked for sign with a second reference (3DGS from all DSLR images). *Tier B:* SIMPLER [35] paired sim/real evaluations of published policies; the twin must reproduce their direction. P = task success rate.

Stage I – C1, VLM-guided task-aware capture (RQ1, H1; sem. 3–4).

- *Method:* the twin's uncertainty, per region (Fisher information [25] or a Bayes' Rays-type [26] field adapted to 3DGS) and per physical parameter (posterior spread), is weighted by task relevance: the VLM's grounding of the instruction, how often expert trajectories pass through a region, and how much expert success changes across a parameter's posterior. The next views or interactions are chosen greedily where task-weighted uncertainty is highest, from the recorded pool (capture time estimated as a tour).
- *Baselines:* uniform capture; task-blind uncertainty selection (FisherRF-type [25], ASID-type [28]); risk-weighted [27] and VLM-guided reconstruction-quality selection (AREA3D-type [29]); same budget.
- *Metric and criterion (H1):* budget–P curves (≥ 4 budgets per testbed; the VLA LoRA-SFT-tuned in each twin); $\geq 40\%$ less capture than uniform at equal P, and the upper 95% bound of $C_{\text{task}} / C_{\text{recon}} < 1$ (§7).



Stage II – C2, uncertainty-aware fine-tuning in the twin and a twin-grounded world model (RQ2, H2; sem. 4).

- *Method*: the VLA is LoRA-fine-tuned by SFT on expert (planner) demonstrations in the twin and then by RL. (a) Randomization strength follows the twin's uncertainty (per-region appearance and geometry, physical parameters from their posterior), and twin samples far from a few real samples in DINOv2 [33] space are down-weighted; the real samples are a held-out slice of the capture and count towards the budget. (b) The world model, post-trained on twin rollouts, generates extra rollouts from states in the twin's most uncertain regions for on-policy RL [20], weighted by that uncertainty.
- *Baselines*: (a) uniform randomization of appearance [31] and dynamics [32], and none; (b) twin only (same C2 without the world model) and world model only; same capture budget.
- *Criterion (H2)*: (a) mean paired P gain over uniform randomization ≥ 10 pp on held-out scenes, lower 95% bound > 0 ; (b) lower 95% bound of the paired gain of twin + world model over twin only > 0 ; both at each of ≥ 2 capture budgets.

Stage III – C3, few real trials and correction (RQ3, H3; end of sem. 4 – sem. 5).

- *Selection*: candidate trials are scored by the predicted gap: the twin's uncertainty along the planned trajectory, twin–world-model disagreement on the outcome and the VLA's uncertainty (spread of LoRA ensemble actions). The top-k are run in reality (tier A: the reference).
- *Use*: the trials correct the twin (re-capture or re-weighting where real and twin observations disagree, counted in the budget; posterior update [7]), extend the world model's post-training data, and fine-tune the VLA by co-training with twin data [14]. Real P from few trials uses prediction-powered inference [34].
- *Baselines*: random selection, uniform coverage and failure-prone selection from the twin alone (TwinRL-type [17]); same number of trials.
- *Criterion (H3)*: the target P with $\geq 50\%$ fewer trials than random (upper 95% bound of the ratio < 1).

Stage IV – the whole method in both testbeds (RQ4, H4; sem. 6).

- *Budget curves*: real data (capture + trials, one operator-time cost) against P for the method; for **real-only fine-tuning of the same VLA** (LoRA SFT on expert demonstrations in the reference, same initialization; real-only RL also shown); and for the **strongest existing twin fine-tuning pipeline for VLAs**, TwinRL/RialTo-style [17, 3]: uniform capture, uniform randomization, SFT + RL of the same VLA in the twin, random or failure-driven real trials (the better is reported), with the same twin, learner and correction and TwinRL's released code where it applies. The exchange rate is the ratio of the budgets reaching the target P.
- *Success criterion (H4, the thesis; §7)*: with C1–C3 and their hyperparameters unchanged, in each testbed $\leq 10\%$ of the real data of real-only fine-tuning (upper 95% bound < 0.2) and $\geq 2\times$ less than the pipeline (bound < 1); H1–H3 effects keep their sign; tier B agrees in direction.

Tier C validation. Two robot campaigns (sem. 5 and 7) with own phone or RGB-D captures at PWr: a mobile robot in 2 rooms and, where access is agreed, a tabletop manipulator; uniform capture versus the method, ~ 8 robot-hours per campaign, agreement only. **Statistics:** tests follow §7; the protocol is pre-registered before each stage. Code is released where dataset and model licences permit.

10. Streszczenie popularnonaukowe / Abstract for general public

Streszczenie popularnonaukowe należy przygotować w dwóch językach: polskim i angielskim. Wersje językowe muszą być tożsame. Opis powinien być napisany językiem popularnonaukowym i zawierać cel rozprawy, opis planowanych badań naukowych oraz powodów, dla których podjęta została ta tematyka badawcza, a także najważniejsze spodziewane efekty (maks. 1 strona, czcionka 11 pt., interlinia 1). / The description for general public must be in Polish and in English. The language versions must be identical. The description should be written for general public and should include the



doctoral dissertation goal, description of research, reasons for attempting a particular research topic and substantial results expected (at most 1 page, font 11, line spacing 1).

Streszczenie popularnonaukowe

Roboty, które mają chwytać przedmioty lub poruszać się po budynkach, coraz częściej korzystają z dużych, wstępnie wytrenowanych modeli sztucznej inteligencji, które rozumieją obraz i polecenia wydane słowami i na ich podstawie wybierają ruchy. Aby taki model dobrze działał w nowym miejscu i przy nowym zadaniu, trzeba go jeszcze nauczyć, a zbieranie prawdziwych danych i prób jest powolne i kosztowne. Dlatego douczanie coraz częściej odbywa się w cyfrowym bliźniaku: wiernej kopii prawdziwego miejsca, którą można odtworzyć z nagrań i pomiarów, łącznie z wyglądem, kształtem i tym, jak rzeczy się poruszają. Powstaje w ten sposób pętla: zbieramy dane w rzeczywistości, budujemy kopię, douczamy w niej model, sprawdzamy go w rzeczywistości, poprawiamy i dopiero wtedy wdramy. Każdy z tych kroków wymaga jednak prawdziwych danych.

Celem rozprawy jest opracowanie metody, która przejdzie przez tę pętlę z dużo mniejszą ilością prawdziwych danych niż dotychczasowe podejścia, niezależnie od zadania. Metoda ma trzy części, po jednej na każdy krok pętli. Zamiast nagrywać wszystko po równo, model rozumiejący obraz i język wskaże, co jest ważne dla zadania, a nagrywać będziemy przede wszystkim te miejsca, o których kopia wie jeszcze za mało. Ponieważ kopia nigdy nie jest idealna, model będziemy douczać tak, aby nie polegał na jej niepewnych fragmentach, a tam, gdzie kopia jest niepewna, dodatkowe sytuacje wygeneruje model świata, czyli program, który uczy się przewidywać, co wydarzy się dalej. Na koniec wybierzemy tylko te nieliczne próby w rzeczywistości, które najwięcej mówią o tym, gdzie system jeszcze zawodzi, i wykorzystamy je do poprawienia kopii, modelu świata i samego modelu. Tę samą metodę sprawdzimy na dwóch różnych zadaniach: chwytaniu przedmiotów ramieniem robota i poruszaniu się robota po budynkach. Badania będą prowadzone przede wszystkim na publicznie dostępnych danych, a wybrane wyniki zostaną potwierdzone na prawdziwym robocie.

Spodziewanym efektem jest jedna metoda, która pozwoli zbierać mniej danych, uczyć skuteczniej i wykonywać mniej kosztownych prób w rzeczywistości, oraz pomiar, ile prawdziwych danych oszczędza w porównaniu z douczaniem wyłącznie na prawdziwych danych i z dotychczasowymi podejściami. Może to przyspieszyć i potęgować wdrażanie robotów w fabrykach, magazynach, szpitalach i biurach. Opracowane oprogramowanie zostanie udostępnione publicznie.

Abstract for general public

Robots that grasp objects or move around buildings increasingly rely on large, pretrained artificial intelligence models that understand images and spoken or written instructions and choose movements from them. For such a model to work well in a new place and on a new task, it still has to be trained further, and collecting real data and trials is slow and costly. That is why this further training increasingly takes place in a digital twin: a faithful copy of a real place that can be rebuilt from recordings and measurements, including how things look, their shape and how they move. This creates a loop: we collect data in reality, build the copy, train the model further in it, check it in reality, correct it and only then deploy it. Each of these steps, however, needs real data.

The goal of this dissertation is to develop a method that goes through this loop with much less real data than existing approaches, whatever the task. The method has three parts, one for each step of the loop. Instead of recording everything evenly, a model that understands images and language will point out what matters for the task, and we will record mainly the places about which the copy still knows too little. Because the copy is never perfect, we will train the model so that it does not rely on the uncertain parts of the copy, and where the copy is uncertain, extra situations will be generated by a world model, a program that learns to predict what happens next. Finally, we will choose only the few trials in reality that tell the most about where the system still fails, and use them to correct the copy, the world model and the model itself. We will test the same method on two different tasks: grasping objects with a robot arm and a robot moving around buildings. The research will be carried out mainly on publicly available data, and selected results will be confirmed on a real robot.



The expected result is one method that makes it possible to collect less data, train more effectively and carry out fewer costly trials in reality, and a measurement of how much real data it saves compared with training on real data only and with existing approaches. This may make deploying robots in factories, warehouses, hospitals and offices faster and cheaper. The software developed will be made publicly available.

11. Termin oddania do druku artykułu naukowego lub monografii / Date of submission for publication of at least one scientific article or monography

Termin wysłania do recenzji artykułu naukowego lub monografii naukowej zgodnie z wymaganiami art. 186 ust. 1 pkt 3 Ustawy stawianymi osobie ubiegającej się stopień doktora. Termin dotyczy oddania do druku co najmniej jednego artykułu naukowego opublikowanego w czasopiśmie naukowym lub w recenzowanych materiałach z konferencji międzynarodowej w odpowiedniej dyscyplinie naukowej, które w roku opublikowania artykułu w ostatecznej formie były ujęte w wykazie („lista ministerialna”) sporządzonym zgodnie z przepisami wydanymi na podstawie Ustawy (art. 267 ust. 2 pkt 2 lit. b) lub 1 monografii naukowej wydanej przez wydawnictwo, które w roku opublikowania monografii w ostatecznej formie było ujęte w wykazie sporządzonym zgodnie z przepisami wydanymi na podstawie Ustawy (art. 267 ust. 2 pkt 2 lit. a). / A date of submission for publication of at least one scientific article published in a scientific journal or in the peer-reviewed materials of an international conference in the relevant scientific discipline, which in the year of publication of the article in its final form were included in the list drawn up in accordance with the regulations issued under the Act (Art. 267 sec. 2 point 2(b) or 1 scientific monograph published by a publishing house, which in the year of publication of the monograph in its final form was included in the list drawn up in accordance with the regulations issued pursuant to the Act (Art. 267 sec. 2 point 2(a), referred to in Art. 186 sec. 1 point 3 of the Act.

May 2027 / maj 2027

Paper **P1** submitted for review to the main track of the **Conference on Neural Information Processing Systems (NeurIPS 2027)**. NeurIPS is peer-reviewed international conference proceedings worth **200 points** on the ministerial list (conference Lp. 87), assigned to the discipline *informatyka techniczna i telekomunikacja*. P1 reports the first two components of the method (Stage I and the first Stage II results, RQ1–RQ2) for a pretrained open vision-language-action model fine-tuned in digital twins: C1, capture of real data guided by a vision-language model to the task-relevant, uncertain parts of the twin (H1), and C2, uncertainty-aware, parameter-efficient fine-tuning in the twin and in a world model grounded in it (first H2 results). Both are evaluated in manipulation and navigation by the real data they save, not by the fine-tuning technique itself (task T4.2). If it is rejected, the revised paper goes to **CVPR 2028** (expected deadline November 2027, also 200 points), and then to the next 200-point conference (§12).

12. Inne / Other comments

Dodatkowe wyjaśnienia, komentarze, podsumowanie związane z indywidualnym planem badawczym (maks. 1 strona, czcionka 11 pt., interlinia 1). / Additional explanations, comments, summary related to the individual research plan (at most 1 page, font 11, line spacing 1).

Planned outputs by year (200-point conferences assigned to the discipline, list of 5.01.2024).

Year	Papers (planned submission, 200 points each)	Code	Grants	Mobility, events
1 (2025/26)	—	—	—	—



2 (2026/27)	P1 (method components C1 and first C2 results, RQ1–RQ2): NeurIPS 2027, May 2027 (§11)	Loop on tier A	SzD Minigrant; NCN PRELUDIUM; NAWA Bekker	Summer school poster
3 (2027/28)	P2 (component C3, few real data, RQ3): ICML or CVPR 2028; P3 (the whole method in both testbeds, RQ4): NeurIPS 2028	C3 code	—	Mid-term; 3-month visit (sem. 6)
4 (2028/29)	Optional P4, consolidated study: ICLR 2029 or CVPR 2029	Method release	—	Dissertation

Foreign research visit (candidate hosts, not yet contacted). (1) Z. Kira's group, Georgia Tech: EmbodiedSplat [5]. (2) Multi-robot Systems Lab, Stanford University (M. Schwager): Gaussian-splatting scene models. A European host (Erasmus+ route) working on active reconstruction or sim-to-real transfer, and a foreign co-author for P2, are sought with the supervisor in sem. 4.

Collaboration at PWr. Robot validation is planned with the K29 *Denali* laboratory (Pioneer 3-DX, Jaguar 4x4, ROS 2) and, if available, a PWr manipulator; a written agreement is task T3.2 (Nov 2026).

Risks and mitigation (semester affected).

- *One component gives no gain* (3–5): its budget curve still answers its RQ; H4 is still tested.
- *Twin uncertainty is poorly calibrated* (3–4): two estimators and an ensemble fallback.
- *Proxy reality is easier than reality* (4–7): separate sources for twin and reference; tiers B and C check the direction (§9).
- *Real-only fine-tuning is strong at small budgets* (6): gains saturate quickly in a known location [13]; H4 is reported as a full budget curve.
- *Compute* (3–7): ~23k H100-hours (§9); WCSS and PLGrid grants in sem. 3; LoRA and small VLAs for development; if short, fewer seeds and one VLA.
- *Crowded field, scooping* (3–6): VLA fine-tuning in simulation and world models is crowded, and twin RL for VLAs exists [17, 21]; we claim only the real-data budget of the loop; early preprints.
- *Licences, robot access* (3, 5, 7): OpenVLA weights carry Llama 2 terms and some checkpoints are gated, so openpi's π_0 is the fallback; without a robot, tier C moves by a semester (tier A decides).
- *Rejection at a top venue* (4–7): each paper has a resubmission path inside the 200-point set 2–4 months later (NeurIPS → CVPR → ICML → ECCV/ICCV).

Prior work (background only). A pre-PhD NLP paper (ACL 2025 SRW, forecasting from hidden states); not part of the dissertation.

Ethics and data. Own captures are anonymized (GDPR); public datasets are used under their licences.

13. Określenie planowanej formy współpracy z promotorem / Planned form of cooperation with the supervisor

Krótki opis planowanej formy współpracy z promotorem, np. częstotliwość spotkań, forma spotkań i ich zakres, sposób komunikacji. / Brief description of the planned form of cooperation with the supervisor, e.g., frequency of meetings, form of meetings and their scope, method of communication.

- **Weekly meeting** (~1 h, in person at K46 or online via MS Teams) to discuss progress, results and next steps.



- **Monthly written progress summary** (completed tasks, results, problems, plan). It lists every IPB task from §3 with its **% completion**, in the same form as the task table of the mid-term self-report.
- **Semester review** against the IPB schedule (§3), before each annual report and the mid-term evaluation. Deviations from the plan and their reasons are recorded and agreed with the supervisor.
- **Supervisor co-signs** the annual reports and the mid-term self-report (autoreferat).
- **Paper-writing sprints** before each submission deadline, with joint editing in Overleaf. The supervisor co-authors and reviews the papers before submission.
- **Ongoing communication** via e-mail / Teams chat. The code and experiment logs are in a shared Git repository that the supervisor can access.

14. Opinia promotora pomocniczego / Opinion of assistant supervisor

Promotor pomocniczy (jeżeli jest wyznaczony) powinien zaopiniować IPB. / Assistant supervisor (if appointed) should provide an opinion of the IRP.

nie dotyczy / not applicable

.....
Podpis promotora pomocniczego /
Signature of assistant supervisor

15. Podpisy doktoranta i promotora/promotorów / Signatures of doctoral student and supervisor(s)

Wybierz datę/Select date

.....
Data / Date

.....
Podpis doktoranta / Signature of doctoral student

.....
Podpis promotora / Signature of supervisor

.....
Podpis drugiego promotora / Signature of second supervisor